Exno: 01 Date: 17/07/2026

Study of various commands used in linux and Windows

Aim:-

To study about various networking commands on Linux operating System.

Basic linux commands.

i) pwd.

description: prints the current working Directory

Output:-

$ pwd
/home/linuxuser.

ii) mkdir

Description: Creates a new directory

output:

mkdir fedora.
ls.
Desktop Documents Downloads fedora.

cd.

Description: changes the current directory

Output:-

cd fedora.
pwd.

/home/[illegible]/fedora.

iv) touch filename.txt

description: Creates an empty file

Output:-

touch example.txt

ls

example.txt

v) echo "Hi" > filename.txt

description: writes text into a file

output:

echo "Hi" > example.txt

vi) cat filename.txt.

description: Displays the contents of a file

output:-

$ cat example.txt.

Hi

vii) grep

description: Searches for a specific pattern in a file.

Output.

grep "H" example.txt.

Hi

viii) wc.

Description: Displays the number of lines, words, and bytes in a file.

Output:-

wc example.txt.

1 1 5 example.txt.

ix) wc -l

Description: Counts the no of lines in a file

Output.

wc -l example.txt.

1 example.txt.

x) wc -w.

Description: Counts the no of words in a file

Output:-

wc -w example.txt.

1 example.txt.

xi) wc -c.

Description: counts the no of bytes in a file

Output:-

wc -c example.txt.

3 example.txt.

xii) wc -m.

Description: counts the

iii) wc - L

Description :- Display the length of the longest line in a file

output :-

wc - L example.txt.

2 example.txt

Basic Networking commands.

i) hostname.

Description : Displays the System's hostname

Output :-

$ hostname

Localhost - Lee

ii) route.

Description : Displays the routing table.

Output.

$ route.

kernal IP routing table.

iii) netstat

Description :- Shows network connections and routing information

Output :-

$ netstat

Active Internet connections.

iv) ip address show.

Description : Displays Ip address and network Interface details.

Output.

ip address show

1 : lo:

2 : enp0s3:

v) nslookup

Description:-

Finds the Ip address of a domain

Output:-

nslookup www.google.com

Name : www.google.com

Address : 142.250.x.xx

vi) ping

Description : Test connectivity to a host.

output:-

ping www.google.com

64 bytes from ....

vii) arp.

Description : Displays the ARP cache.

Output:-

arp -a

- gateway (10.0.2.2).

viii) traceroute.

Description : Displays the route taken by packets to reach a destination

output:-

traceroute www.google.com

1 - gateway

2. ...

Result:-

The basic Linux and networking commands were executed Smoothly and expected outputs were obtained.

Expno: 02.
Date: 23/07/2026

Learning and Assignment of Ip Address Manually to Computers

Aim:-

To learn Ip addressing and manually assign an Ip address to a computer. To verify the assigned Ipaddress using linux Networking Commands.

Theory:-

An IP (Internet portocal) address is a unique numerical address assigned to every device connected to a network. It enables communication between devices over a network. An IPV4 address consists of four octets seperated by dots (eg. 192.168.1.10)

An Ip address can be assigned either manually (Static Ip) or automatically (Dynamic Ip) using DHCP. An manually assigned Ip address remains fixed unless changed by the user. Along with the Ip address, parameters such as the Subnet mask, default Gateway and DNS server are also configured for proper Network Communication.

Commands and Sample outputs

i) Display all IP Addresses

Command

Ip address show

Sample output

2: ens160:
inet 172.16.211.128/24
inet 6.
fe80::20c:29ff:fe75:9927/64.

Observation: Displays the IPV4 and IPV6 addresses of all network Interfaces.

ii) Display a Specific network Interfaces command.

ip addr show dev ens160.

Sample output:

2: ens160:
inet 172.16.211.128/24.

Observation: Displays the Ip Address assigned to the Specified Interfaces.

iii) Display Routing table.

Command.

ip route show.

Sample output.

default via 172.16.211.2 dev ens160
172.16.211.0/24 dev ens160

Observation: Displays the routing table and the default gateway.

ii) Display Neighbour (ARP) Table.

Command.

Ip neigh show.

Sample output.

172.16.211.1 REACHABLE

172.16.211.254 STALE

Observations: Displays the neighbouring devices and their MAC Addresses.

Manual Ip Configuration.

parameter	value.
IP Address	192.168.1.10
Subnet Mask.	255.255.255.0.
Default Gateway	192.168.1.1.
DNS server.	8.8.8.8.

Result:-

The Ip address was manually assigned and verified successfully using Linux networking commands.

Exno: 03

Date: 23/07/2026.

Study of Different Types of Network Cables and Crimping of Cable with RJ45 Connection

Aim:-

To study different types of network cables and understand the crimping process using an RJ45 connector.

Concept:-

Network Cables are used to connect computers and other networking devices for data communication. The most commonly used cable is LAN is the unshielded Twisted pair (UTP) Cable An RJ45 connection is attached to the ends of the cable of established a network connection. The wires are arranged according to the T568A or T568B wiring standard before crimping and the completed cable is tested to ensure proper connectivity.

Types of Network Cables:-

Cable types	Description
i) UTP (Unshielded Twisted pair)	Commonly used for Ethernet LANs and office networks
ii) STP (Shielded Twisted pair)	Used in areas with high electrical or electromagnetic interference.

iii) Coaxial cable – Used for cable television broadband internal and CCTV systems.

iv) Optical Fiber Cable – Used for high-speed high Bandwidth and long distance communication.

Crimping Procedure:-

1). Cut the UTP Cable to the required length.
2). Strip about 2-3cm of the outer insulation
3) Untwist and straighten the wires.
4) Arrange the wires allording to the T568B
5) Trim the wire ends evenly.
6) Insert the wires into the RJ45 Connector.
7) Crimp the connector using a crimpt tool.
8) Test the cable using a LAN cable tester

Observations.

* The wires were arranged accordingly to the T568B standard.
* The RJ45 connector was crimped successfully.
* The Cable tester indicated proper continuity
* The cable was ready for network. Communication.

Result:

The different types of Network Cables were studied and the UTP Cables was successfully terminated using the RJ45 Connection and tested for proper Connectivity.

Exp no: 04.

Date: 06/08/2026.

IMPLEMENT PACKET SNIFFING USING RAW SOCKETS IN PYTHON.

Aim:-

To implement packet sniffing using raw sockets in Python and capture network packets passing through the network interfaces.

Theory:-

* Packet Sniffing is the process of monitoring and capturing data packets travelling across the network.

* A packet sniffer intercepts packets and analyzes their content.

* Raw Sockets provide direct access to lower level network protocols, allowing applications to capture packets before they are processed by operating system.

* The technique is used for network priority, trouble shooting and security analysis.

Algorithm:-

i) Import the required Socket module

ii) Create a new Socket

iii) Bind the Socket to the network interface

iv) Receive packets using the recvform() method

v) Display the Captured packet Information

vi) Repeat the process continuously to monitor network traffic.

Program:-

```
Import Socket
Sniffer = Socket.Socket(Socket.AF_INET, Socket.
Sock.Raw.Socket.IPPROTO_IP)
host = Socket.gethostbyname(Socket.gethostname())
Sniffer.bind((host,0))
Sniffer.SetSocket(Socket.IPPROTO_IP, Socket.
IP_HDRINCL,1)
```

while true:-

raw_data, addr = sniffer.recvfrom(65535)

print("packet received from:", addr).

print (raw_data).

Output:-

packet received from:-

('192.168.1.5', 0)

b' E\x00\x00\x00 ....

packet received from:-

('192.168.1.10', 0)

b' E\x00\x00\x00 ....

Result:-

Thus, packet sniffing was successfully implemented using raw sockets in python & network packet were captured and displayed.

Exp no: 05
Date: 24/7/2026

Develop customized ping commands to Test Server Connectivity

Aim:-

To develop and execute customized ping commands in windows to test server connectivity measure network response time, and analyze packet transmission statistics.

Introduction

The ping command is a built-in windows utility used to test communication between a computer and another device over an IP network. It works by sending ICMP (Internet Control message protocol) Echo Requests packets to the destination and receiving Echo reply packets.

i) Why do we use the ping command.

The ping command is used to test network connectivity between two devices, verify server availability, measure response time, detect packet loss and troubleshoot network related problems.

ii) What is the purpose of ping command.

The ping command is a network diagnostic tool used to verify connectivity between a computer and another device (server, router or website) over an IP network.

3. Ping Commands

i) ping google.com.

Description

Test basic Connectivity to the server.

Output.

Reply from 142.250.67.46 : bytes = 32 times = 13ms TTL = 118.

ping statists from 142.250.67.46

packets : Sent = 4, Received = 4. lost = 0 (0% loss)

Approximate round trip times in milliseconds:

min = 6ms, max = 16ms, avg = 13ms

ii) ping -l 1024 google.com.

Description :-

Sends packets with a size of 1024 bytes.

Output :-

useage : ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS], [-r count] [-s count] [[-j host-list] | [-k host-list]] [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p] [-4] [-6] targetname.

iii) ping -w 2000 google.com.

Description.

Sets the timeout to 2000 milliseconds

output:-
pinging google.com [142.250.67.46] with 32 bytes of data.

ping statistics for 142.250.67.46:
Packets: Sent = 4, Received = 4, Loss = 0 (0% loss),
Approximate round trip times in mille-seconds:
Minimum = 5ms, maximum = 6ms, Avg = 5ms

iv) ping -t google.com.

Description
continously pings the server until stopped using ctrl+c.

output:-
pinging google.com [142.250.67.46] with 32 bytes of data.

Request timed out
ping statistics for 142.250.67.46:
Packets: Sent = 27, Received = 26, Lost = 1 (3% loss),
Approximate round trip times in mille-seconds.
Min = 4ms, max = 32ms, Avg = 9ms
Control-c.

v) ping 8.8.8.8.

Description:-
Test connectivity using google's public DNS Ip address

Output:-

Pinging 8.8.8.8 with 32 bytes of data:

ping statistics for 8.8.8.8:-

packets: Sent=4, Received: 4, lost=0(0% loss),

Approximate round trip times in milli seconds:-

min=6ms, max=18ms, Avg=11ms

vi) ping localhost

Description

Test the local machine's network stack.

output:-

Pinging DESKTOP-3T5ALVB [::1] with 32 bytes of data:

Ping statistics for ::1:

packets: Sent=4, Received=4, lost=0 (0% loss),

Approximate round trip times in milli seconds:

min=0ms, max=0ms, Avg=0ms

Result:-

The customized ping commands were executed successfully on windows system. Different parameters such as packet count, size, timeout, continuous ping, Ip address were tested.

Expno: 09.
Date: 13/08/2025

Design a simple topology & configure with one Router. Two Switches and PCs using Cisco packet Tracer.

Aim:-

To design and configure a simple network topology consisting of one router, two switches and multiple PCs in Cisco packet Tracer, and enable communication between PCs.

Definitions

* Router: A networking device that forwards data packets between computer networks, directing traffic on the internet
* Switch: A device that connects devices within a network and uses MAC addresses to forward data to the correct destination.
* PC (personal Computer): End devices in a network that users interact with to send & receive data.
* Cables:-
  * Straight-through Cable: Used to connect different devices like PC to Switch or Switch to router.
  * Cross over Cable: Used to connect similar devices, like Switch to Switch. (rarely needed with modern Switches).
  * Console cable: Used to connect PC to router or switch Console port for configuration

Equipment Needed.

* Cisco Packet Tracer software
* 1 Router (eg.. 2811)
* 2 switches (eg... 2960)
* 4 PCs.
* Appropriate cables (Copper Straight-Through)

Step by Step procedure

Step 1:- Open cisco packet Tracer.

* Launch Cisco packet Tracer on your computer

Step 2: Add devices to workspace.

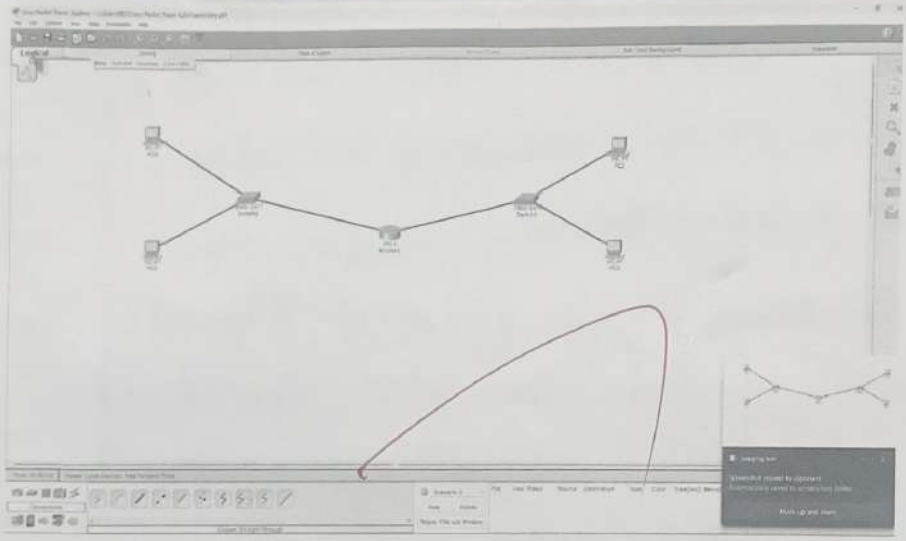
* From the bottom device toolbar, drag and drop
  * 1 Router (1941)
  * 2 switches (2960)
  * 4 PCs (2 connected to each switch)

Step 3: Connect devices with cables

* Use copper Straight-through cables to connect
* Router Gigabit Ethernet 0/0 to switch 0 Fast Ethernet 0/1
* Router Gigabit Ethernet 0/1 to switch 1 Fast Ethernet 0/1
* PCs to the switches
* PC 0 to switch 0 Fast Ethernet 0/2
* PC 1 to switch 0 Fast Ethernet 0/3

* PC2 to switch1 FastEthernet 0/2.

* PC3 to switch1 FastEthernet 0/3.



memory.

steps: · Configure PC Ip Addresses.

* Click on each PC > Desktop tab > Ip Configuration.

and assign the following Ip's and gateways:-

Device	Ip Address	Subnet Mask	Default gateway
PC0	192.168.1.10	255.255.255.0	192.168.1.1
PC1	192.168.1.11	255.255.255.0	192.168.1.1
PC2	192.168.2.10	255.255.255.0	192.168.2.1
PC3	192.168.2.11	255.255.255.0	192.168.2.1

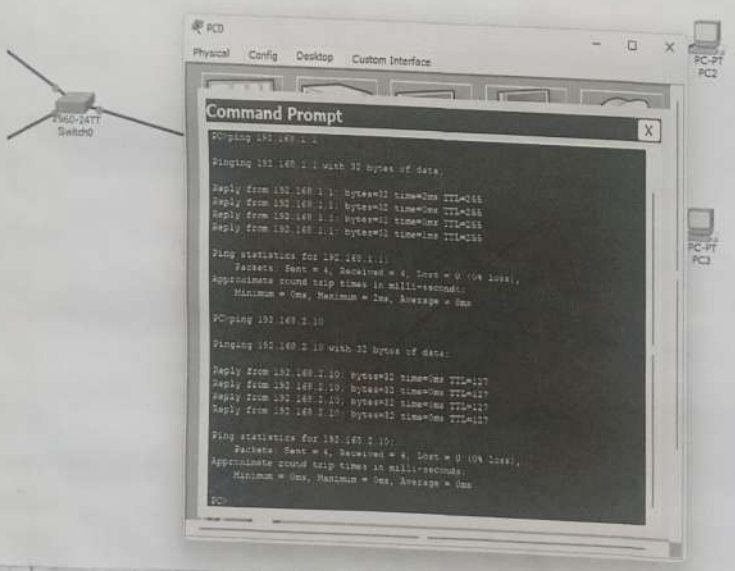
Step 6:- Save & test Connectivity.

* Save the packet tracer file.

* Test connectivity using the ping command from PC0 to PC2 or PC3:-

* Go to PC0 > Command prompt > type ping 192.168.2.10.

* You should see replies if the network is correctly configured.



Result:-

The PCs connected to different switches can communicate with each other via the router. The successful ping replies verify that the network is properly configured and operational.

Exp no: 10.
Date:- 13/02/2026.

Customize switch with network modules using Cisco packet Tracer.

Aim:-

To learn how to add a network module to a Cisco switch, connect multiple PCs and a server, assign IP addresses, and test connectivity in Cisco Packet Tracer.

Definition

A network module is an additional component added to a switch to provide more ports or connection types HWIC-4ESW is used in this experiment.

procedure:-

i) Open Cisco packet tracer and setup yours work space

ii) Drag and drop a 2960-2477 switch to canvas

iii) Add five PC's and one Server-PT from the End Devices panel.

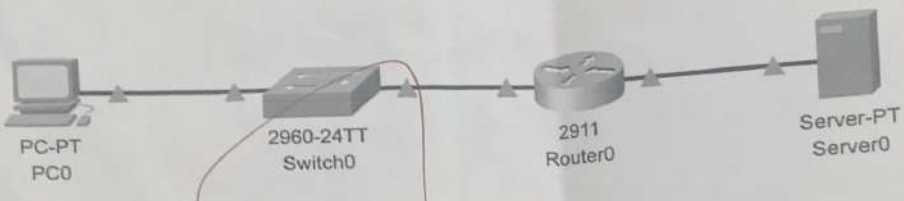
iv) power off the switch:-

Click the switch, open the physical tab, and turn off all the power button

v) Install HWIC – 4ESW module on the switch

vi) power on the switch. and let it boot

vii) Connect all



i) PC0 : 192.168.1.10

ii) PC1 : 192.168.1.11

iii) PC2 : 192.168.1.12

iv) All Subnet Masks : 255.255.255.0

Test Connectivity ∈ from any PC, open Command prompt and ping other devices).

Inputs :-

Devices :

→ 1 Cisco 2960-24TT Switch

* 1 HWIC → 2+ESW network module.
* 5 pcs (pc0 - pc4)
* 1 server - PT (Server 0)

Cables :-

* Copper Straight - through cables Ip configuration.
* Use static Ip address as listed above.

Output :-

A working network setup in which all pcs and the server can successfully communicate. Confirming that the built-in and module ports are functioning.

Result :-

The switch was successfully customized with a network module, and all connected devices communicated successfully.

Exp no : 07

Date :- 20/08/2026

DEVELOP A SIMPLE CALCULATOR USING XMLRPC

Aim :-

TO develop a simple Calculator application using the XML-RPC protocol in python that performs basic arithmetic operations like addition, subtraction, multiplication and division through client-server communication.

Description :-

XML-RPC (Extensible Markup Language Remote Procedure Call) is a protocol that uses xml to encode requests and responses and HTTP as a transport mechanism. It enables communication between a client & server where the client invokes methods hosted on the server remotely.

In this experiment, a simple calculator service is created using XML-RPC in python:

* The server hosts arithmetic functions
* The client connects to the server to perform operations like addition, subtraction, multiplication and division.

Software Requirements :-

* Operating System : Windows / Linux
* Python version : 3.x or above
* Modules Required : xmlrpc.server, xmlrpc.client (built in with Python).
* Text Editor or IDE (VS code / PyCharm)

Procedure :-

i) Open a Python IDE or terminal

ii) Create two separate python files : server.py and client.py.

iii) Write the provided server side code in server.py and client side code in client.py.

iv) Run the server.py file first to start the XML RPC server.

v) Run the client.py file to perform arithmetic operations remotely.

vi) Observe the results displayed on the client screen and verify the server outputs.

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\dell\OneDrive\Desktop\CN> python client.py
XML-RPC Calculator
------------------
Enter first number: 45
Enter second number: 8

Choose an operation:
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice: 3
Result: 360.0
PS C:\Users\dell\OneDrive\Desktop\CN>
```

```
def multiply(a,b):
    return a*b
def divide(a,b):
    if b==0:
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\dell\OneDrive\Desktop\CN> python client.py
XML-RPC Calculator
------------------
Enter first number: 450
Enter second number: 7

Choose an operation:
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice: 4
Result: 64.28571428571429
PS C:\Users\dell\OneDrive\Desktop\CN>
```

```
server.register_function(divide,
server.serve_forever()
```

client.py.

```
import xmlrpc.client

proxy = xmlrpc.client.ServerProxy("http://localhost:8000/")

print("Addition: ", proxy.add(10,5))
print("Subtraction: ", ...
```

```
PS C:\Users\dell\OneDrive\Desktop\CN> python client.py
XML-RPC Calculator
------------------
Enter first number: 10
Enter second number: 30

Choose an operation:
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice: 1
Result: 40.0
PS C:\Users\dell\OneDrive\Desktop\CN>
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\dell\OneDrive\Desktop\CN> python client.py
XML-RPC Calculator
------------------
Enter first number: 60
Enter second number: 27

Choose an operation:
1. Addition
2. Subtraction
3. Multiplication
4. Division
Enter your choice: 2
Result: 33.0
PS C:\Users\dell\OneDrive\Desktop\CN>
```

The XML-RPC ... successfully implemented. The client communicated with the server using XML-RPC, performed all arithmetic operations correctly and displayed accurate result

Exp No: 12

DATE: 24.9.2026

NMAP to discover live host using ARP scan, Icmp scan and tcp/Udp ping scan

Aim:

To discover live host in a network using Nmap with ARP, ICMP, TCP, and UDP ping scans using Try hack Me platform

Definition:

Nmap host discovery is used to identify active / live hosts in a network. It uses different probes such as ARP, ICMP, TCP and UDP to determine whether a host is reachable

ARP - Address resolution protocol

. ARP is used to find the MAC address of a device from its IP address within local network

ICMP -> Internet control message protocol

. ICMP is a network protocol used for error reporting and network protocol diagnostics. The ping command commonly uses ICMP Echo and Echo reply messages

TCP - Transmission control protocol

TCP is a connection-oriented transport protocol that provides reliable and ordered delivery of data between devices

UDP → User Datagram Protocol

UDP is a connectionless transport protocol that sends data like host establishing a connection It is faster but doesn't guaranteed delivery

Nmap:

Nmap is a network scanning tool used to discover live hosts, identify open ports, gather information about network services

Procedure:

[i] open the assigned tryHackme room and start the target machine

[ii] open the terminal provided in the tryHackme environment

[iii] Note the target IP address provided by tryHackMe

[iv] perform ARP, ICMP, TCP and NDP host discovery scan using Nmap

[v] observe the output and identify the hosts shown as "Host is up"

Nmap commands used

[i] General live Host discovery :-

command :nmap - Pn 10.49.151.126

observation :- The target host was detected as alive

[ii] ARP - Scan :-

command :- nmap :- sn - PR 10.49.151.126

observation : The ARP scan successfully detected the target as a live host

[iii] ICMP Scan :

command: nmap -sn - PE 10.49.151.126

observation : The ICMP scan detected the target as live host.

[iv] TCP - SYN Ping :

command: nmap -sn - PS 10.49.151.126

observation : The TCP SYN scan detected the target as a live host

[v] TCP ACK Ping

command: nmap sn - PA 10.49.151.126

observation :- The TCP ACK scan detected the target as a live host

[vii] UDP Ping:

Command: nmap -sn -PU 10.0.4.151.126

Observation:- The UDP scan detected the target as a live host

Result:

The discovery of live active host in a network using nmap by performing ARP scan I can use TCP/IP Ping scan in Try Hack Me Platform verified Successfully

Ex No: 6

DATE.

Building Anonymous FTP scanner using FTPLAB MODULE

AIM:

To create a python based FTP scanner using the ftplib module that:

* Checks if a public FTP server allows anonymous login.
* Lists the contents of directories on the server.

PROCEDURE:

* Import the built-in ftplib module.
* Write two python scripts:
* Sender side: Connects to the FTP server, logs in anonymously and lists the root directory.
* Receiver side: Connects anonymously and lists a specific sub-directory.
* Run the scripts in Google Colab or any Python environment.
* Observe the login status, server message, and directory contents.

CODE:

Vendor side code:

```
ftp lib
def anonymous login (host name):
    try:
        ftp = ftp lib.FTP (hostname)
            = ftp.login
        anonymous login

    Print (response)

    If "230" in response:
        + str (hostname) + "FTP Anonymous Login succ")
        Print ("welcome message: \n", ftp.getwelcome
        ())

        Print ("[-]" + str (hostname) + "FTP.
        Anonymous failed") Print ("Error", str(e))

        ftp 'ls.debian.org'
        anonymous login (hostname)
```

Receiver side code:

```
import ftp lib
    def check - anonymous - connection (host, path):
    try:
        with ftp.lib.FTP (host) as connection:
        connection.login ('anonymous', 'anonymous')
        Print (f"Listing directory: {path}:")
        connection.cwd (path) for name, details
        in connection.mlsd ():
```

- Print(name, details['type'], details.get('size', 'N/A'))

except Exception as e:

Print("Error", str(e))

FTP_server_URL = 'ftp.cse.cdlu.org'.

DOWNLOAD_PATH '/pub/linux/kernel/v5.x/'

check_anonymous_connection(FTP_server_URL, download_dir_path)

OUTPUT: Server side output:

230 Anonymous access granted, restrictions apply [+] ftp.cse.cdlu.org FTP-Anonymous Login Successful welcome message:

220 PRO FTPD Server (mirror.cse.cdlu.org) [::ffff:192.168.1.4]

Client side output:

drwxrwxr-x ftp ftp

Received data output:

Welcome: 220 PRO FTPD Server (mirror.cse.cdlu.org [::ffff:192.168.1.4])

Listing directory: /pub/linux/kernel/v5.x

linux-5.10.1.tar.xz file 113072504

linux-5.10.11.tar.xz file 113167464

RESULT:

* the script successfully connected to FTP server
* It verified anonymous login is allowed
* The directory contents were retrieved and displayed
* The Proves that the server provides public access and is working correctly

Ex No: 11

Date

Examine Network Address Translation using Cisco Packet Tracer.

Aim:

Implement Network Address translation using Cisco packet tracer will configure static dynamic and port Address translation on a router to enable communication between a private internal and a public external network

Procedure:

1. open Cisco packet tracer
2. Add devices:

   Drag PC0 and PC1 representing inside private network
3. connect devices:

Step 1: open Cisco Packet Tracer

Step 2: Add devices

- Add switches
- Add routers
- Add a cloud or Another Router

Step 3:

1. Connect PC1 and PC2 to the switch:

* Select copper straight-through cable & click on PC1, choose the FastEthernet port

2. Connect switch to Router

3. Connect Router to outside Network device

Step 4: Verify Connection

* The links should show green or orange when devices are connected properly

* PC1: 192.168.1.10/24

* PC2: 192.168.1.11/24

* Router GigabitEthernet 0/0: 192.168.1.1/24 outside Network:

* Router GigabitEthernet 0/1: 200.1.1.1/24 show of Router

Step 2: Assign IP Addresses

Step 1: Assign IP Address to PC1 and PC2

* Click on PC1 in the workspace

* Go to desktop tab

* Click on IP Configuration

* Enter the following

* IP 192.168.1.10 Subnet Mask 255.255.255.0

- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

2. Assign IP Address to Router interfaces

- click on the Router.
- click on the CLI tab

using the config tab

- Go to GigabitEthernet 0/0
- Set the IP Address: 192.168.1.1
- Subnet Mask: 255.255.255.0
- turn on the interface by checking Port status

do on

- Go to GigabitEthernet 0/1
- set the IP Address 200.1.1.1
- Subnet mask: 255.255.255.0
- Turn on the interface by checking Port status

do on using the CLI method:

enable configure terminal

3. Assign IP to outside device:

- If it is a Router
- IP to it a cloud
- click Home Go config tab
- Assign the IP address 200.1.1.2 with

Subnet mask 255.255.255.0

Step 4: Verify - IP configuration

A From PC1 and PC2 go to the desktop tab →
command prompt :.

A ping the router inside interface :

A From Router IP. any the outside device ping

200.1.1.2 step 3: open the router CLI and enter the following commands :

CODE:

Step 4: enable

configure

terminal

interface gigabit ethernet 0/0 ip

address 192.168.1.1 255.255.255.0 no shutdown

exit

interface gigabit ethernet 0/1 ip address

200.1.1.1 255.255.255.0

no shutdown

exit

4.1 configure inside and outside interfaces for

: configure terminal interface

gigabit ethernet 0/0 ip nat

inside exit interface.

gigabit ethernet 0/1 ip nat

outside exit

4.2 Static NAT configuration

4.3 Dynamic NAT Configuration

How to set Default Gateway on PC1 and PC2

1. Click on PC1 in the workspace
2. Go to the desktop tab
3. Click on IP configuration.
4. On the Default Gateway field type 192.168.1.1
5. Make sure the IP Address and Subnet mask are already set.
6. Close the window

6.1 Check the NAT translation

6.2 Test connectivity

RESULT:

The router successfully translated private IPs to public ones using static, dynamic, and PAT configuration

All inside host could communicate with the external network, confirming that NAT worked properly in packet tracer

Ex No. 13
Date:

Demonstrate Network Forensics using pcapxray tool

Aim:

To analyze and visualize network traffic from a PCAP file using the pcapxray tool for forensic investigation.

Steps to complete network forensics Experiment using pcapxray

Step 1: Prepare your computer

Make sure you have Python 3 installed

If you don't have Python:

Go to python.org and download it

Follow the installation instructions for your operating system

Step 2: Install pcapxray tool

1. Open Command Prompt or Terminal.
2. Type this command and press Enter
3. pip install pcapxray

This installs the pcapxray tool on your computer

Step 3: Get a PCAP file

Option A: Use a sample PCAP file.

Download a sample PCAP file from a website like

https://wiki.wireshark.org/SampleCaptures

Save the file on your computer

Option B: Capture your own traffic.

Open Wireshark.

Click on the network interface to start capturing.

Let it capture for 1-2 minutes.

Step 4: Run pcap xray on your PCAP file.

1. Put the PCAP file in a folder you can easily access.

2. Open command prompt terminal and navigate to that folder. Example:

3. cd path to your folder

4. Run the command:

5. pcap xray.py -f capture.pcap

Step 5: View the result:

After the command runs, it will create a dir named pcapxray.

Step 6: Analyze what you see

Look for the biggest nodes - these are the most active devices

RESULT:

Pcap Xray successfully generated a visual map of hosts & connections and protocols, helping identify suspicious or abnormal network activities

Ex No:4

Date:

TO CAPTURE, SAVE, AND ANALYZE NETWORK TRAFFIC ON TCP / UDP / IP / HTTP / ARP / DHCP / ICMP, DNS using wireshark tool

Aim:

The aim of the experiment is to capture, save and analyze network traffic on TCP / UDP / IP / HTTP / ARP / DHCP / ICMP / DNS using wireshark tool.

procedure:

Step 1: Download and install wireshark from its official website. ensure you have administrator privileges for network capture tool.

Step 2: Launch wireshark and identify the network interface through which your device is connected to the network.

Step 3: Before starting the capture, apply capture filters to focus on specific protocols and reduce unwanted data.

Example of capture filters you can use:

Capture only TCP traffic : tcp

Capture only UDP traffic : udp

Capture only HTTP traffic : tcp port 80

Capture only DNS traffic : dns

Capture only ARP traffic : arp

Capture DHCP traffic : port 67 or port 68

Step 4: Click the start button to begin capturing the network

Step 5: While capturing use display filters to isolate and analyze specific protocols more easily. Example,

To display only UDP packets: udp

Step 6: Let the capture run until you have collected enough packets

Step 7: Stop the captured traffic by navigating

Step 8: Review and analyze the captured packets in detail. Look for key information

output:

The output will be list of network packets displayed in wireshark's packet details window which will include.

Sample output:

No.	Time	Source	Destination	Protocol	
1	0.00000	192.168.1.1	8.8.8.8	ICMP	
2	0.002748	8.8.8.8	192.168.1.1	ICMP	64 Echo Reply
3	0.003736	192.168.1.1	192.168.1.1		

RESULT:

Thus the experiment capture done and analyze network traffic on TCP/UDP/IP/HTTP/DNS

Expt No:
Date:

Analyze the different types of servers using
- Webalizer TOOL

AIM:

To analyze the usage statistics of different type of servers using webalizer tool

PROCEDURE:

1. Install the webalizer tool in the system
Sudo apt-get update sudo
apt-get install webalizer
2. verify installation
3. Locate the log files of the server.
4. Run webalizer on the log file to Generate a report.
5. Open a web browser and open the report
6. Repeat the same process
7. Compare and analyze the reports

PROGRAM:

Step 1: Install webalizer sudo apt-get install webalizer

Step 2: Run webalizer

/var/log/apache2/access.log.0/var/www/html/apache_report/

Step 3: Run webalizer for Nginx logs sudo webalizer

/var/log/nginx/access.log.0.

Step 4: Run Webalizer for proxy logs sudo webalizer

./var/log/squid/access.log.0

Sample observation Table

- Server
- Type
- Total
- Hits
- Unique
- Visitors
- Bandwidth
- Used
- Top

Client / Domain Report URL

Apache 12,560 1240 1.8 GB 192.168.1.10.

Pie chart of top hosts and URLs

RESULT:

The webalizer tool successfully generated. Graphical and statistical reports of different servers Total hits and visits

Bandwidth usage

TOP IP's and URL's

Daily monthly and hourly traffic

Exp No: 8

Date

Develop a program to create reverse shell using TCP socket

Aim:

To develop a simple chatting application using Python socket programming to exchange messages between a client and a server.

Observation

Chatting allow users to exchange message through a computer network.

Procedure:

1. Install Python and open VS code
2. Create two files server.py and client.py
3. Import the socket and threading modules
4. Create a TCP socket in both program
5. Bind and start the server, then connect the client
6. Run the server first and the client next
7. Exchange messages between the two terminals
8. Type exit to end the chat.

Program:

Server program:

```
import socket
import threading

server = socket.socket()
server.bind(("127.0.0.1", 12345))
server.listen()
print("waiting for client...")
conn, addr = server.accept()

def receive():
    while true:
        msg = conn.recv(1024).decode()
        if not msg or msg.lower()=="exit"
        print("client", msg)

threading.Thread(target = receive, daemon
                                =true).start()

while true:
    msg = input("server:")
    conn.sendall(msg.encode())
    if msg.lower == 'exit':
        break

conn.close()
server.close()
```

Client Side:

```
client = socket.socket()
client.connect(("127.0.0.1", 12345))
print("Connected to server")
def received
    while True:
        msg = client.recv(1024).decode()
        if not msg or msg.lower() == "exit":
        print("server:", msg)

while True:
    msg = input("client: ")
    client.sendall(msg.encode())
    if msg.lower() == "exit":
        break
client.close()
```

output:

Server:	Client:
waiting for client	connected to server:
client connected	client: Hi
server: Hello	Server: Hello
client: Hi	client: I am fine
server: How are you?	server: How are you
client: I am fine	

Result :

This a Simple chatting application was implemented successfully using Python socket programming and TCP to exchange message between the client and server.